Worksheet 34 · Trigonometry IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use sine, cosine and tangent to find missing sides • Use inverse trigonometric functions to find missing angles • Solve problems involving angles of elevation and depression		
Grade: / 58	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Trigonometry*

Question 1 (5 marks)

The diagram shows a right-angled triangle.

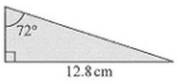


Diagram **NOT** accurately drawn

Five of these triangles are put together to make a shape.

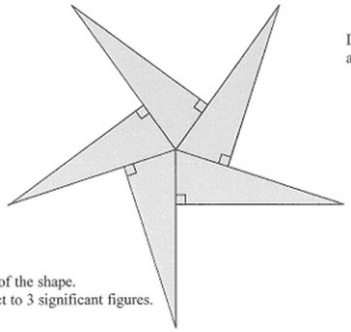


Diagram **NOT** accurately drawn

Calculate the perimeter of the shape.
Give your answer correct to 3 significant figures.

..... cm

Question 2 (3 marks)

The diagram shows triangle *ABC*

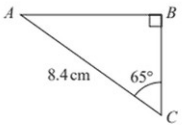


Diagram **NOT** accurately drawn

Work out the length of the side *AB*
Give your answer correct to 3 significant figures.

..... cm

Question 3 (3 marks)

Here is a right-angled triangle.

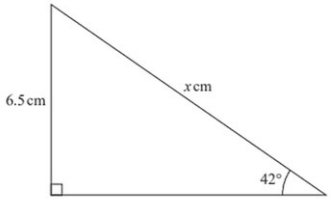


Diagram **NOT** accurately drawn

Work out the value of *x*.
Give your answer correct to one decimal place.

x = ...

Question 4 (3 marks)

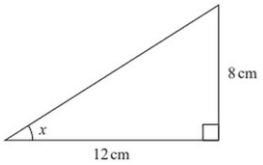


Diagram **NOT** accurately drawn

Calculate the size of angle *x*.
Give your answer correct to 1 decimal place.

.....

Question 5 (4 marks)

Here is isosceles triangle ABC.

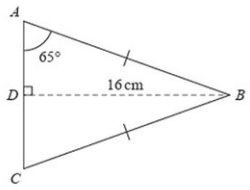


Diagram **NOT** accurately drawn

D is the midpoint of AC and $DB = 16$ cm.

Angle $DAB = 65°$

Work out the perimeter of triangle ABC.
Give your answer correct to one decimal place.

..... cm

Question 6 (5 marks)

In the diagram, ABC is a right-angled triangle and DEF is a semicircular arc.

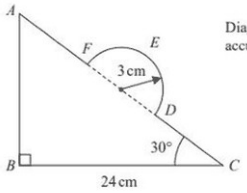


Diagram **NOT** accurately drawn

In triangle ABC

$BC = 24$ cm angle $ABC = 90°$ angle $BCA = 30°$

The points D and F lie on AC so that DF is the diameter of the semicircular arc DEF
The radius of the semicircular arc is 3 cm.

Work out the length of $AFEDC$
Give your answer correct to 2 significant figures.

..... cm

Question 7 (3 marks)

The diagram shows two vertical phone masts, AB and CD, on horizontal ground.



Diagram **NOT** accurately drawn

$AB = 6.2$ m $AC = 244$ m $CD = 30.7$ m

Work out the size of the angle of depression of B from D
Give your answer correct to one decimal place.

..... °

Question 8 (3 marks)

The diagram shows two hot air balloons.
A is a point on the base of one of the balloons and B is a point on the base of the other balloon.

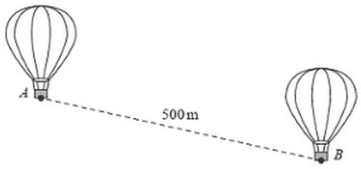


Diagram **NOT** accurately drawn

The distance between A and B is 500 metres.
The angle of depression of B from A is 23°

Calculate the vertical height of A above B.
Give your answer correct to one decimal place.

..... metres

(Total for Question 13 is 3 marks)

Question 9 (6 marks)

The diagram shows a vertical cliff with a vertical radio mast on top of the cliff and a buoy in the sea.

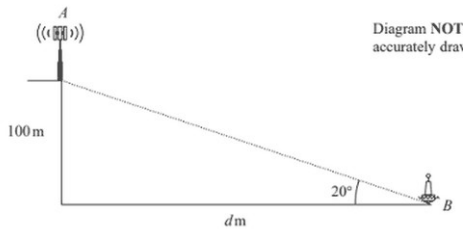


Diagram **NOT** accurately drawn

The height of the cliff is 100 metres.
The buoy is at the point B that is d metres from the base of the cliff.
The angle of elevation from B to the top of the cliff is 20°

(a) Calculate the value of d.
Give your answer correct to 3 significant figures.

$d =$ .....
(3)

The point A at the top of the radio mast is vertically above the top of the cliff.
The angle of elevation from B to A is 25°

(b) Calculate the height of the radio mast.
Give your answer correct to 3 significant figures.

..... m
(3)

Question 10 (4 marks)

The diagram shows two right-angled triangles, DEF and EFG.

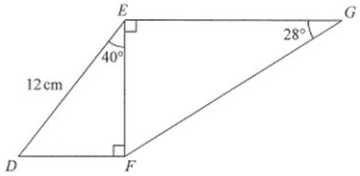


Diagram **NOT** accurately drawn

Work out the length of EG.
Give your answer correct to 3 significant figures.

Question 11 (3 marks)

A zip wire is shown as the dashed line AC in the diagram.

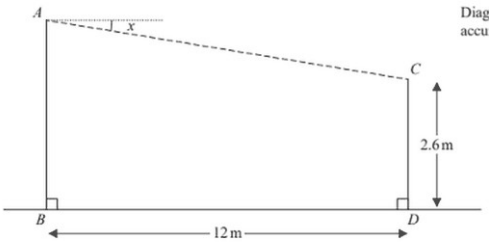


Diagram **NOT** accurately drawn

The zip wire is supported by two vertical posts AB and CD standing on horizontal ground.

$CD = 2.6\,\text{m}$ $BD = 12\,\text{m}$

The zip wire makes an angle x with the horizontal, as shown in the diagram.
The design of the zip wire requires the angle x to be at least 5°

Work out the least possible height of the post AB
Give your answer correct to 3 significant figures.

..... m

Question 12 (3 marks)

Here is a right-angled triangle.

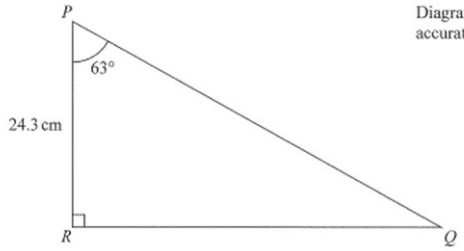


Diagram **NOT** accurately drawn

Calculate the length of PQ.
Give your answer correct to 3 significant figures.

Question 13 (4 marks)

Here is triangle ABD.

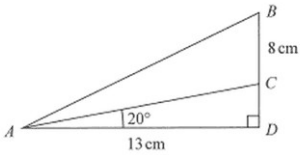


Diagram **NOT** accurately drawn

The point C lies on BD.

$AD = 13\,\text{cm}$ $BC = 8\,\text{cm}$ angle $ADB = 90°$ angle $CAD = 20°$

Calculate the size of angle BAC.
Give your answer correct to 1 decimal place.

Question 14 (5 marks)

$ABCD$ is a trapezium.

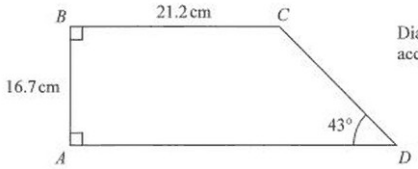


Diagram **NOT** accurately drawn

Calculate the perimeter of the trapezium.
Give your answer correct to 3 significant figures.

Question 15 (4 marks)

ABC is an isosceles triangle with $BA = BC$.

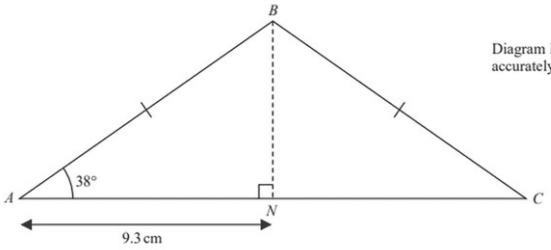


Diagram **NOT** accurately drawn

N is the point on AC such that $AN = 9.3$ cm and BN is perpendicular to AC.

Work out the perimeter of triangle ABC.
Give your answer correct to 3 significant figures.